

```
[1]: import re
import pandas as pd

[2]: text = """
Contact us at support@example.com or admin@university.edu.
Visit https://www.example.com, http://nlp.org or https://github.com.
Phone numbers are +91-9876543210, +91 8765432109 and 020-12345678.
Important dates are 15/08/2026, 01-01-2027 and 25/12/2026.
Follow #ArtificialIntelligence, #MachineLearning, #NLP and #DeepLearning.
"""

[3]: # Email Pattern
email_pattern = r'\b[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}\b'

# URL Pattern
url_pattern = r'https?://[^\s]+'

# Phone Number Pattern
phone_pattern = r'(?:\+91[- ]?)?[0-9]{10}|[0-9]{3}-[0-9]{7}'

# Date Pattern
date_pattern = r'\b\d{1,2}[/-]\d{1,2}[/-]\d{4}\b'

# Hashtag Pattern
hashtag_pattern = r'#\w+'

[5]: emails = re.findall(email_pattern, text)
urls = re.findall(url_pattern, text)
phones = re.findall(phone_pattern, text)
dates = re.findall(date_pattern, text)
hashtags = re.findall(hashtag_pattern, text)

print("Emails:", emails)
print()
print("URLs:", urls)
print()
print("Phone Numbers:", phones)
print()
print("Dates:", dates)
print()
print("Hashtags:", hashtags)

Emails: ['support@example.com', 'admin@university.edu']

URLs: ['https://www.example.com,', 'http://nlp.org', 'https://github.com.']

Phone Numbers: ['+91-9876543210', '+91 8765432109', '020-12345678']

Dates: ['15/08/2026', '01-01-2027', '25/12/2026']

Hashtags: ['#ArtificialIntelligence', '#MachineLearning', '#NLP', '#DeepLearning']

[6]: max_len = max(
    len(emails),
    len(urls),
    len(phones),
    len(dates),
    len(hashtags)
)

emails += [''] * (max_len - len(emails))
urls += [''] * (max_len - len(urls))
phones += [''] * (max_len - len(phones))
dates += [''] * (max_len - len(dates))
hashtags += [''] * (max_len - len(hashtags))

df = pd.DataFrame({
    "Emails": emails,
    "URLs": urls,
    "Phone Numbers": phones,
    "Dates": dates,
    "Hashtags": hashtags
})
```

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))  
df
```

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[6]:
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	Emails	URLs	Phone Numbers	Dates	Hashtags
0	support@example.com	https://www.example.com,	+91-9876543210	15/08/2026	#ArtificialIntelligence
1	admin@university.edu	http://nlp.org	+91 8765432109	01-01-2027	#MachineLearning
2		https://github.com.	020-1234567	25/12/2026	#NLP
3					#DeepLearning

```
[9]: def validate_email(email):  
    pattern = r'^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$'  
    return bool(re.match(pattern, email))  
  
def validate_url(url):  
    pattern = r'^https?://[^\s]+$'  
    return bool(re.match(pattern, url))  
  
def validate_phone(phone):  
    pattern = r'^(?:\+91[- ]?)?[0-9]{10}$|^[0-9]{3}-[0-9]{7}$'  
    return bool(re.match(pattern, phone))  
  
def validate_date(date):  
    pattern = r'^\d{1,2}[/-]\d{1,2}[/-]\d{4}$'  
    return bool(re.match(pattern, date))
```

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[10]: print("Email Validation")  
  
for email in emails:  
    if email:  
        print(email, "->", validate_email(email))  
  
print("\nURL Validation")  
  
for url in urls:  
    if url:  
        print(url, "->", validate_url(url))  
  
print("\nPhone Number Validation")  
  
for phone in phones:  
    if phone:  
        print(phone, "->", validate_phone(phone))  
  
print("\nDate Validation")  
  
for date in dates:  
    if date:  
        print(date, "->", validate_date(date))
```

Email Validation
support@example.com -> True
admin@university.edu -> True

URL Validation
https://www.example.com, -> True
http://nlp.org -> True
https://github.com. -> True

Phone Number Validation
+91-9876543210 -> True
+91 8765432109 -> True
020-1234567 -> True

Date Validation
15/08/2026 -> True
01-01-2027 -> True
25/12/2026 -> True

```
[ ]:
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